02 — Requirements Documentation (SRS)

Prepared 9 November 2026 for teacher review on 10 November. **Draft; not approved or implemented.**

Source: 03-software-requirements-specification.md

Software requirements specification

Version: 0.1 · **Prepared:** 9 November 2026 · **Status:** Proposed review draft; no teacher approval or product test execution claimed.

1. Purpose and authority

This SRS is a navigation and baseline-control document for the proposed release. It composes the linked requirement, role, rule and acceptance registers without creating alternative requirement IDs. Teacher acceptance, campus authorization, implementation and tests are separate decisions/evidence.

2. Product perspective

ClassroomCast comprises a local browser interface, application backend, SFU/signaling services, policy manager, approved enforcement integration and persistent operational records. Campus infrastructure ownership and browser capture consent remain external constraints.

3. Normative candidate sections

Topic	Authoritative location in this package
Original numbered capabilities and infrastructure obligations	Functional register
Quality, offline and security constraints	Non-functional requirements
Role and resource boundaries	Permissions
State and concurrency rules	Business rules
Actor goals	Use cases
Acceptance evidence	Test catalogue
Requirement/story/test connections	Traceability

Topic	Authoritative location in this package
Unresolved choices and changes	Decision log

4. Definitions

SYS = System Administrator; TCH = Teacher; STU = Student; NET = IT/Network Administrator. SFU = Selective Forwarding Unit: receives publication and forwards media to permitted viewers. Signaling exchanges connection/session information; it is distinct from the media stream. Policy manager validates and orchestrates policies; enforcement service applies supported rules. T = Teacher requirements; S = Student requirements; NAC = Network Access Control; CN = Campus Network; OR = teacher-owned router / portable-network requirements. These are original requirement families; F denotes workflow, UC user goal, CC delivery item, US story, TC test specification, D unresolved decision.

5. Assumptions and dependencies

Supported laptop browsers, trusted local endpoints, reachable SFU/media paths, authorized network integration and usable server/AP capacity must be established by feasibility tests. No cloud runtime dependency is assumed. External websites require WAN; offline media does not prove internet availability or selective restriction effectiveness.

6. Baseline/change procedure

Before implementation commitment, agree test conditions for the specified 50–150 students and equipment for both deployment modes, sign off supported browsers and acceptance targets, and assign decision owners. For changes record requested text, affected IDs, effort/schedule/risk/test impact, decision maker, date and evidence. Preserve supplied Must requirements until an explicit change is approved. Approval fields remain blank; revision 0.1 is not a signed baseline.

Source: 04-functional-requirements.md

Functional requirements and deployment obligations

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1. Source classification

The 63 rows retain original IDs and source priorities: 31 Must, 7 Should, 3 Could, 22 individually unlabeled deployment obligations. CN/OR are not optional merely because priorities were unlabeled. Their proposed Must* classification needs confirmation. Capability availability is not established by documentation.

2. Requirement register

ID	Requirement	Source priority	Flows	Use cases
NAC01	Enable or disable network restrictions for a class	Must	F01 F10	UC06
NAC02	Block website categories, including AI and coding agents	Must	F09 F10 F11	UC05 UC06
NAC03	Block or allow individual domains	Must	F09 F10 F11	UC05 UC06
NAC04	Support allow-by-default and restricted-by-default policies	Must	F09 F10 F11	UC05 UC06
NAC05	Automatically apply policies when a class starts	Must	F01 F10	UC06
NAC06	Restore the previous policy when a class ends	Must	F13 F14	UC08 UC09
NAC07	Save reusable policy profiles	Must	F09 F10 F11	UC05 UC06

ID	Requirement	Source priority	Flows	Use cases
NAC08	Add allowlist exceptions for approved websites	Must	F09 F10 F11	UC05 UC06
NAC09	Schedule restrictions by date and time	Should	F17	UC12
NAC10	Display active policy and actual enforcement status	Must	F10 F13 F14 F15	UC06 UC08 UC09 UC10
NAC11	Record administrator actions and timestamps	Must	F10 F13 F14 F15	UC06 UC08 UC09 UC10
NAC12	Provide an emergency disable and rollback function	Must	F14 F15	UC09 UC10
NAC13	Support temporary rules with automatic expiry	Must	F14 F15	UC09 UC10
NAC14	Import and export policy configurations	Should	F17	UC12
NAC15	Apply different policies to student groups or devices where supported	Could	F18	UC13
NAC16	Restrict policy management to authorized administrators	Must	F09 F10 F13 F14 F15	UC05 UC06 UC08 UC09 UC10
NAC17	Keep router credentials and API secrets on the server	Must	F09 F10 F13 F14 F15	UC05 UC06 UC08 UC09 UC10
NAC18	Prevent restrictions from breaking ClassroomCast itself	Must	F09 F10 F13 F14 F15	UC05 UC06 UC08 UC09 UC10
NAC19	Minimize student browsing-history collection	Must	F09 F10 F13 F14 F15	UC05 UC06 UC08 UC09 UC10
NAC20	Support locally stored rules for offline operation	Should	F03 F17	UC12 UC15
T01	Create a classroom with a unique session ID and join code	Must	F04 F05	UC01 UC02

ID	Requirement	Source priority	Flows	Use cases
T02	Start, stop, pause, and resume broadcasting	Must	F06 F07	UC03
T03	Share an entire screen, window, or browser tab where supported	Must	F06 F07	UC03
T04	Share supported system audio and optional microphone audio	Must	F06 F07	UC03
T05	View connected students and their connection status	Must	F05 F08	UC02 UC04
T06	Approve, reject, remove, and block participants	Must	F05 F08	UC02 UC04
T07	Configure resolution, frame rate, and bitrate	Must	F06 F07	UC03
T08	Display the classroom URL, join code, and QR code	Must	F04 F05	UC01 UC02
T09	Display streaming health and connection diagnostics	Should	F15	UC10
T10	Share presentation files, PDFs, images, and videos	Should	F16	UC11
T11	Display session history and participant counts	Could	F18	UC13
S01	Join through a local URL or QR code	Must	F05	UC02
S02	Enter a classroom join code	Must	F05	UC02
S03	Watch the teacher's screen in real time	Must	F06 F07	UC03
S04	Play supported shared audio with volume control	Must	F06 F07	UC03
S05	View the broadcast in fullscreen	Must	F06 F07	UC03
S06	Reconnect after temporary network interruptions	Must	F12	UC07
S07	Show connection status and useful errors	Must	F05 F07 F12 F15	UC02 UC03 UC07 UC10
S08	Enter a student name or assigned student ID	Should	F05	UC02
S09	Download teacher-provided materials when permitted	Should	F16	UC11
S10	Raise a hand, submit questions, or answer polls	Could	F18	UC13

ID	Requirement	Source priority	Flows	Use cases
CN01	Use an approved local server address or hostname	Unlabeled / proposed Must*	F02	UC14
CN02	Verify student devices can reach the application and signaling services	Unlabeled / proposed Must*	F02	UC14
CN03	Confirm required WebRTC media traffic is permitted	Unlabeled / proposed Must*	F02	UC14
CN04	Verify Wi-Fi client isolation and guest-network restrictions	Unlabeled / proposed Must*	F02	UC14
CN05	Obtain campus IT approval for hosting and firewall changes	Unlabeled / proposed Must*	F02	UC14
CN06	Validate Wi-Fi capacity and network throughput for 150 students	Unlabeled / proposed Must*	F02	UC14
CN07	Provide QR-code and manual URL discovery	Unlabeled / proposed Must*	F02	UC14
CN08	Avoid IP address conflicts with existing campus services	Unlabeled / proposed Must*	F02	UC14
CN09	Host required application components locally for offline operation	Unlabeled / proposed Must*	F02	UC14
CN10	Integrate network restrictions with campus-approved DNS or gateway controls	Unlabeled / proposed Must*	F02	UC14
OR01	Business-class Wi-Fi 6 or better equipment suitable for high client density	Unlabeled / proposed Must*	F03	UC15
OR02	Configurable classroom SSID and Wi-Fi password	Unlabeled / proposed Must*	F03	UC15
OR03	WPA2-AES or WPA3 security where compatible	Unlabeled / proposed Must*	F03	UC15

ID	Requirement	Source priority	Flows	Use cases
OR04	DHCP capacity for all students and support devices	Unlabeled / proposed Must*	F03	UC15
OR05	Stable server IP address or local hostname	Unlabeled / proposed Must*	F03	UC15
OR06	Wired Gigabit Ethernet or faster connection to the server where possible	Unlabeled / proposed Must*	F03	UC15
OR07	Appropriate client isolation settings that permit access to the server and approved enforcement services	Unlabeled / proposed Must*	F03	UC15
OR08	One or more access points based on classroom capacity testing	Unlabeled / proposed Must*	F03	UC15
OR09	Support for additional access points and a wired switch	Unlabeled / proposed Must*	F03	UC15
OR10	Portable power, cables, and carrying equipment	Unlabeled / proposed Must*	F03	UC15
OR11	Fully functional local services without WAN or internet access	Unlabeled / proposed Must*	F03	UC15
OR12	Local software-based policy management without requiring manual rule changes for each class	Unlabeled / proposed Must*	F03	UC15

3. Supporting features and count reconciliation

Card	Capability	Basis	Priority/status
BF-05-07	Basic health and operations	Shared architectural and classroom-administration obligations; no extra original ID	REQUIRED / source obligation
BF-06-05	Classroom authorization	Unnumbered classroom-administration obligation; supports T01/T06 and S03	REQUIRED / source obligation
BF-06-06	Application account management	Proposed account model, not a separate original numbered feature	PROPOSED / no source priority
BF-06-07	Role assignment	Proposed RBAC allocation	PROPOSED / no source priority

Card	Capability	Basis	Priority/status
BF-06-08	Application settings and scoped review	Proposed administration UI/permission details	PROPOSED / no source priority

All 68 original board cards are available in board/ROW-2-FEATURE-CARDS.md. They are grouped feature descriptions; user stories and tasks may split or combine them. The traceability matrix maps every numbered requirement to a story and test. No feature has been marked implemented.

Source: 05-non-functional-requirements.md

Non-functional requirements

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1. Quality obligations

Local runtime, SFU delivery, trusted capture context, server-side secrets, truthful enforcement status, health/logging/recovery and minimal data collection come from supplied architectural obligations. Performance numbers below remain proposals. These constraints apply across selected features, including optional ones once included.

2. Candidate measurable conditions

Test basis	Proposed setup/target	Approval/evidence rule
AT01 — Local UI	Fresh browser reaches join page within 5 s on a reachable healthy test LAN; 10 trials per supported browser	Timing starts at navigation and ends when code input is interactive; record cold/cache conditions
AT02 — Playback quality	Agreed starting profile 720p/15 fps; source-to-display p95 ≤1 s; report received bitrate, freezes, packet loss and per-client results	Proposed threshold needs D12 approval; measure visible source timestamp at source/viewer in a common recording; separate screen content from encoder adaptation
AT03 — Capacity/stability	50/100/150 concurrent subscribers, 30 min each; per-viewer playback active at ≥98% of declared one-minute checks; no open release-blocking defect	Record physical/simulated counts, AP/SSID/topology, OS/browser, source, bitrate, server CPU/RAM/NIC and samples; simulation does not prove 150-laptop Wi-Fi airtime
AT04 — Policy apply	Healthy reachable authorized adapter reports verified outcome within 30 s of class-start/apply event in 10 trials	Must include client-path denied/allowed controls and local app/media access; upstream conditions recorded
AT05 — Restoration	Prior policy verified within 30 s of end/emergency/expiry trigger in 10 healthy reachable trials	In unreachable/conflicting-state tests, class still closes; durable pending/failure shown within proposed 5 s of detecting failure; safe retry/operator recovery required

Test basis	Proposed setup/target	Approval/evidence rule
AT06 — Reconnect	After a 10 s interruption, same-class playback recovers within 15 s of reachability returning in ≥9/10 attempts	Also test class ended and participant revoked while offline; those clients must not regain access
AT07 — Offline	Preprovision runtime assets/images, names/cert trust and lists; disconnect WAN and open fresh browser	External public-site failure is not evidence of selective blocking; use controlled local domains for offline rule checks

Use the teacher-specified 50–150 students, including CN06 validation for 150. D12 defines test conditions and quality thresholds. Lower-load exploratory stages do not discharge the 150-student target; simulated subscribers do not prove physical Wi-Fi capacity.

3. Cross-cutting acceptance

Criterion/test	Proposed acceptance
AC-AUTH / TC-AUTH	Matrix allow/deny and cross-class/scope tests pass; stale/forged/revoked room grants fail; role removal affects new privileged requests; no permission is inferred only from visible UI
AC-LOCAL / TC-LOCAL	AT07 confirms local app, API, signaling, SFU, policy storage and required filtering service operation without cloud/CDN runtime dependence
AC-SFU / TC-SFU	Documented transport path shows teacher publication to local SFU and per-viewer subscriptions; actual negotiated streams/bitrates and load are recorded
AC-RECOVERY / TC-RECOVERY	Execute isolated scenarios TC-FAIL01–TC-FAIL08; health distinguishes network, application, media, trust/name and enforcement faults; unfinished temporary policy operations resume safely; backup/database/config restoration drill documented
AC-PRIVACY / TC-PRIVACY	Server secrets absent from clients/exports/logs; retained participant/audit data matches approved minimal collection and deletion rules

4. Operational quality and unknowns

Availability/SLA, exact hardware sizing, supported browser versions, data retention, backup frequency, recovery point/time objectives and retry ceilings are not confirmed. Measure feasibility and decide with the teacher/network owner; do not invent numeric service guarantees. Record these decisions under D09–D13 and D18–D20.

Source: 07-business-rules.md

Business rules

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1. Proposed rules derived from existing requirements and reviewed workflows

Rule	Required behavior / proposed mechanism	References
BR01	Authorization checks action, class ownership, network scope and current state; default deny	NAC16, TC-AUTH
BR02	Join code required; a valid code does not imply admission, identity or media grant	S02, D02/D03
BR03	Final admission and seat allocation are atomic; recheck pending state, block, code, capacity and class state	F05/F08, TC-ADM05/06
BR04	Publishing may start with zero viewers; current admitted viewers alone receive media	F06, S03
BR05	Pause/stop sharing does not end a classroom or revoke student admission	T02, F07
BR06	Operation IDs bind requester, class, target/publication and action; same completed request returns recorded outcome; unfinished requests resume remaining work	F07/F08/F13
BR07	Recheck authorization/state and publication version before SFU mutation and when responses arrive; stale responses cannot revive closed or replaced publications	F07/F12/F13
BR08	Remove/block revoke grants and prevent refresh; verify actual media denial separately; unknown cleanup schedules recovery	T06, F08/F15
BR09	Saved profile, accepted apply request and verified enforcement are separate states	NAC07/NAC10, F09/F10

Rule	Required behavior / proposed mechanism	References
BR10	Preserve prior policy and operation record before changes; restore only owned changes and reconcile conflicts	NAC06/NAC12/NAC13
BR11	A class is ended only when admission/grant closure and media termination or effective denial are verified; policy restoration is a separate result	F13
BR12	Expiry survives browser close and backend restart; unreachable services produce persistent unresolved status	F14/F15, D11
BR13	Local classroom paths remain protected; domain filtering does not imply universal device control	NAC18, F11
BR14	Evidence distinguishes configured/requested from observed/verified state	NAC10; all recovery flows

2. Proposed state dimensions

Dimension	Candidate states	Important distinction
Classroom	Created, Open, Ending, Ended	Paused media does not close class
Participant	Pending, Admitted, Rejected, Removed, Blocked	Connection loss is not removal
Publication	Not publishing, Publishing, Broadcasting, Paused, Stopping, Stopped, Unresolved	New capture after source ends requires browser consent via F06
Policy operation	Not applied, Applying, Verified, Degraded, Restoring, Restored, Unresolved	Record scope, version, evidence and time

These names are design proposals, not established database enums. Each transition needs validation, durable outcome and appropriate notification. Final D08/D11 choices can refine them.

3. Closure outcome selection

First check application/media closure. If unverified: **Class ending — cleanup unresolved**, regardless of policy result. If verified: report **Class ended** plus restoration **not required**, **verified**, or **pending/failed**. Never label an unknown restore as not required. An unauthorized end request leaves class and publication unchanged.

Source: traceability.md

Requirements, stories, delivery items and tests

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1. Traceability policy

Every original requirement has one requirement-level story candidate and original test specification. CC rows are inherited delivery work, not new requirements. All rows: criteria approval Pending; implementation Not assessed; tests Not run; acceptance Pending. Coverage links do not prove test execution. Supporting architectural and administration capabilities use cross-cutting tests and source feature-card references.

2. Complete matrix

Requirement	Source priority	Story	Flows	Use cases	Delivery items	Test
NAC01	Must	US-NAC01	F01 F10	UC06	CC-012, CC-013, CC-016, CC-018	TC-NAC01
NAC02	Must	US-NAC02	F09 F10 F11	UC05 UC06	CC-011, CC-013, CC-016, CC-018	TC-NAC02
NAC03	Must	US-NAC03	F09 F10 F11	UC05 UC06	CC-011, CC-013, CC-016, CC-018	TC-NAC03
NAC04	Must	US-NAC04	F09 F10 F11	UC05 UC06	CC-011, CC-013, CC-016, CC-018	TC-NAC04
NAC05	Must	US-NAC05	F01 F10	UC06	CC-014, CC-016, CC-018	TC-NAC05
NAC06	Must	US-NAC06	F13 F14	UC08 UC09	CC-014, CC-015, CC-016, CC-018, CC-020	TC-NAC06
NAC07	Must	US-NAC07	F09 F10 F11	UC05 UC06	CC-011, CC-013, CC-018	TC-NAC07
NAC08	Must	US-NAC08	F09 F10 F11	UC05 UC06	CC-011, CC-013, CC-016, CC-018	TC-NAC08
NAC09	Should	US-NAC09	F17	UC12	CC-024	TC-NAC09
NAC10	Must	US-NAC10	F10 F13 F14 F15	UC06 UC08 UC09 UC10	CC-012, CC-013, CC-016, CC-018, CC-020	TC-NAC10

Requirement	Source priority	Story	Flows	Use cases	Delivery items	Test
NAC11	Must	US-NAC11	F10 F13 F14 F15	UC06 UC08 UC09 UC10	CC-014, CC-016, CC-018	TC-NAC11
NAC12	Must	US-NAC12	F14 F15	UC09 UC10	CC-015, CC-016, CC-018, CC-020	TC-NAC12
NAC13	Must	US-NAC13	F14 F15	UC09 UC10	CC-015, CC-016, CC-018, CC-020	TC-NAC13
NAC14	Should	US-NAC14	F17	UC12	CC-024	TC-NAC14
NAC15	Could	US-NAC15	F18	UC13	CC-026	TC-NAC15
NAC16	Must	US-NAC16	F09 F10 F13 F14 F15	UC05 UC06 UC08 UC09 UC10	CC-007, CC-014, CC-016, CC-018, CC-020	TC-NAC16
NAC17	Must	US-NAC17	F09 F10 F13 F14 F15	UC05 UC06 UC08 UC09 UC10	CC-012, CC-016, CC-018, CC-020	TC-NAC17
NAC18	Must	US-NAC18	F09 F10 F13 F14 F15	UC05 UC06 UC08 UC09 UC10	CC-011, CC-015, CC-016, CC-018, CC-020	TC-NAC18
NAC19	Must	US-NAC19	F09 F10 F13 F14 F15	UC05 UC06 UC08 UC09 UC10	CC-014, CC-016, CC-018, CC-020	TC-NAC19
NAC20	Should	US-NAC20	F03 F17	UC12 UC15	CC-023	TC-NAC20
T01	Must	US-T01	F04 F05	UC01 UC02	CC-003, CC-004, CC-018	TC-T01
T02	Must	US-T02	F06 F07	UC03	CC-008, CC-010, CC-018	TC-T02
T03	Must	US-T03	F06 F07	UC03	CC-005, CC-008, CC-018	TC-T03
T04	Must	US-T04	F06 F07	UC03	CC-008, CC-018	TC-T04
T05	Must	US-T05	F05 F08	UC02 UC04	CC-007, CC-010, CC-018	TC-T05
T06	Must	US-T06	F05 F08	UC02 UC04	CC-007, CC-010, CC-018	TC-T06
T07	Must	US-T07	F06 F07	UC03	CC-008, CC-010, CC-018	TC-T07
T08	Must	US-T08	F04 F05	UC01 UC02	CC-003, CC-004, CC-018	TC-T08
T09	Should	US-T09	F15	UC10	CC-023	TC-T09
T10	Should	US-T10	F16	UC11	CC-025	TC-T10
T11	Could	US-T11	F18	UC13	CC-026	TC-T11

Requirement	Source priority	Story	Flows	Use cases	Delivery items	Test
S01	Must	US-S01	F05	UC02	CC-003, CC-006, CC-018	TC-S01
S02	Must	US-S02	F05	UC02	CC-003, CC-006, CC-018	TC-S02
S03	Must	US-S03	F06 F07	UC03	CC-005, CC-006, CC-018	TC-S03
S04	Must	US-S04	F06 F07	UC03	CC-009, CC-018	TC-S04
S05	Must	US-S05	F06 F07	UC03	CC-009, CC-018	TC-S05
S06	Must	US-S06	F12	UC07	CC-009, CC-018	TC-S06
S07	Must	US-S07	F05 F07 F12 F15	UC02 UC03 UC07 UC10	CC-006, CC-009, CC-018	TC-S07
S08	Should	US-S08	F05	UC02	CC-025	TC-S08
S09	Should	US-S09	F16	UC11	CC-025	TC-S09
S10	Could	US-S10	F18	UC13	CC-026	TC-S10
CN01	Unlabeled / proposed Must*	US-CN01	F02	UC14	CC-002, CC-017	TC-CN01
CN02	Unlabeled / proposed Must*	US-CN02	F02	UC14	CC-002, CC-017	TC-CN02
CN03	Unlabeled / proposed Must*	US-CN03	F02	UC14	CC-002, CC-017	TC-CN03
CN04	Unlabeled / proposed Must*	US-CN04	F02	UC14	CC-002, CC-017	TC-CN04
CN05	Unlabeled / proposed Must*	US-CN05	F02	UC14	CC-002, CC-017	TC-CN05
CN06	Unlabeled / proposed Must*	US-CN06	F02	UC14	CC-019	TC-CN06
CN07	Unlabeled / proposed Must*	US-CN07	F02	UC14	CC-002, CC-017	TC-CN07
CN08	Unlabeled / proposed Must*	US-CN08	F02	UC14	CC-002, CC-017	TC-CN08
CN09	Unlabeled / proposed Must*	US-CN09	F02	UC14	CC-002, CC-017, CC-020	TC-CN09
CN10	Unlabeled / proposed Must*	US-CN10	F02	UC14	CC-012, CC-017	TC-CN10
OR01	Unlabeled / proposed Must*	US-OR01	F03	UC15	CC-019	TC-OR01

Requirement	Source priority	Story	Flows	Use cases	Delivery items	Test
OR02	Unlabeled / proposed Must*	US-OR02	F03	UC15	CC-002, CC-017	TC-OR02
OR03	Unlabeled / proposed Must*	US-OR03	F03	UC15	CC-002, CC-017	TC-OR03
OR04	Unlabeled / proposed Must*	US-OR04	F03	UC15	CC-002, CC-017	TC-OR04
OR05	Unlabeled / proposed Must*	US-OR05	F03	UC15	CC-002, CC-017	TC-OR05
OR06	Unlabeled / proposed Must*	US-OR06	F03	UC15	CC-002, CC-017	TC-OR06
OR07	Unlabeled / proposed Must*	US-OR07	F03	UC15	CC-002, CC-017	TC-OR07
OR08	Unlabeled / proposed Must*	US-OR08	F03	UC15	CC-019	TC-OR08
OR09	Unlabeled / proposed Must*	US-OR09	F03	UC15	CC-019	TC-OR09
OR10	Unlabeled / proposed Must*	US-OR10	F03	UC15	CC-002, CC-017	TC-OR10
OR11	Unlabeled / proposed Must*	US-OR11	F03	UC15	CC-002, CC-017, CC-020	TC-OR11
OR12	Unlabeled / proposed Must*	US-OR12	F03	UC15	CC-012, CC-017	TC-OR12

3. Cross-cutting coverage

TC-AUTH, TC-LOCAL, TC-SFU, TC-RECOVERY and TC-PRIVACY cover shared architectural obligations; TC-ADM01–07 and TC-FAIL01–08 supplement admission/fault cases; proposed TC-CTRL01–04 and TC-CLOSE01–03 cover reviewed control/closure races. See test catalogue. Five supporting cards remain identified in functional register; do not invent original source IDs for them.